

# SUMS OF POWERS OF INTEGERS: A COMPLETE FRAMEWORK FOR CLOSED FORMULAS

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ABSTRACT.

## CONTENTS

|                                                                      |    |
|----------------------------------------------------------------------|----|
| 1. Definitions and notations                                         | 2  |
| 2. Historical context                                                | 2  |
| 3. Auxiliary lemmas                                                  | 2  |
| 4. Ordinary sums of powers                                           | 4  |
| 5. Remark on central factorial numbers of the second kind            | 6  |
| 6. Generalized central factorial numbers of the second kind          | 8  |
| 7. Double and triple sums of powers in terms of central differences  | 11 |
| 8. Multifold sums of powers in terms of central differences          | 12 |
| 9. Multifold sums of powers in generalized central factorial numbers | 13 |
| 10. Multifold sums of powers in binomial form                        | 15 |
| 11. Related works                                                    | 17 |
| 12. Framework for sums of powers                                     | 17 |
| 13. Conclusions                                                      | 17 |
| References                                                           | 17 |
| 14. Mathematica programs                                             | 18 |

## 1. DEFINITIONS AND NOTATIONS

Allow us to define the mathematical objects we utilize throughout the manuscript. For multifold sums of powers, we use the notation provided by Donald Knuth in [1], because it is simple and beautiful.

**Definition 1.1** (Multifold sums of powers). *For non-negative integers  $r, n, m$*

$$\begin{aligned}\Sigma^0 n^m &= n^m, \\ \Sigma^1 n^m &= \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m, \\ \Sigma^{r+1} n^m &= \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.\end{aligned}$$

We utilize the following notation for central factorials

**Definition 1.2** (Central factorials). *For integers  $n, k$*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n \left(n + \frac{k}{2} - 1\right) \left(n + \frac{k}{2} - 2\right) \cdots \left(n - \frac{k}{2} + 1\right) = n \left(n + \frac{k}{2} - 1\right)_{k-1}, & \text{if } k > 0, \end{cases}$$

where  $\left(n + \frac{k}{2} - 1\right)_{k-1} = \prod_{j=1}^{k-1} \left(n + \frac{k}{2} - j\right)$  are falling factorials.

J. F. Steffensen gives quite a good discussion on factorial notations and definitions in [2].

## 2. HISTORICAL CONTEXT

## 3. AUXILIARY LEMMAS

**Lemma 3.1** (Central Newton's formula). *For a function  $f$  defined on integers and arbitrary integers  $x, t$*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t),$$

where  $\delta^k f(t)$  denotes the  $k$ -th central finite difference

$$\delta^k f(t) = \sum_{j=0}^k (-1)^j \binom{k}{j} f\left(t + \frac{k}{2} - j\right),$$

and  $(x - t)^{[k]}$  are central factorials defined by (1.2).

*Proof.* See [3, p. 462], and [4, section 2, eq. (19)], and [5, section 6, eq. (21)].  $\square$

**Lemma 3.2** (Newton's formula for powers). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$n^m = \sum_{k=0}^m \frac{(n - t)^{[k]}}{k!} \delta^k t^m.$$

**Lemma 3.3** (Central factorials binomial form). *For integers  $n$  and  $k \geq 1$ ,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k - 1}.$$

*Proof.* We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!} \left( n + \frac{k}{2} - 1 \right)_{k-1} = \frac{n}{k(k-1)!} \left( n + \frac{k}{2} - 1 \right)_{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k - 1},$$

because of the identity in falling factorials  $\frac{(x)_n}{n!} = \binom{x}{n}$ .  $\square$

**Lemma 3.4** (Central factorials binomial form decomposition). *For integers  $n, k \geq 0$ ,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k - 1} = \frac{1}{2} \binom{n + \frac{k}{2}}{k} + \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k}.$$

*Proof.* We have  $\binom{a}{k} = \frac{a}{k} \binom{a-1}{k-1}$ . Let  $a = n + \frac{k}{2}$ , then

$$\binom{n + \frac{k}{2}}{k} = \frac{n + \frac{k}{2}}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} + \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k-1}.$$

Thus,

$$\frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \binom{n + \frac{k}{2}}{k} - \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k-1}.$$

By moving under common denominator, and applying binomial recurrence yields

$$\frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{1}{2} \left[ 2 \binom{n + \frac{k}{2}}{k} - \binom{n + \frac{k}{2} - 1}{k-1} \right] = \frac{1}{2} \left[ \binom{n + \frac{k}{2}}{k} + \binom{n + \frac{k}{2} - 1}{k} + \binom{n + \frac{k}{2} - 1}{k-1} - \binom{n + \frac{k}{2} - 1}{k-1} \right].$$

Thus, the statement follows.  $\square$

**Lemma 3.5** (Newton's formula for powers decomposition). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$n^m = \frac{1}{2} \sum_{k=0}^m \left[ \binom{n-t+\frac{k}{2}}{k} + \binom{n-t+\frac{k}{2}-1}{k} \right] \delta^k t^m.$$

**Lemma 3.6** (Generalized hockey-stick identity). *For integers  $a, b$  and  $j$ ,*

$$\sum_{k=a}^b \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

*Proof.* We have,  $\sum_{k=a}^b \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left( \sum_{k=0}^b \binom{k}{j} \right) - \left( \sum_{k=0}^{a-1} \binom{k}{j} \right)$ . By hockey stick identity  $\sum_{k=0}^n \binom{k}{j} = \binom{n+1}{j+1}$  yields,

$$\sum_{k=a}^b \binom{k}{j} = \left( \sum_{k=0}^b \binom{k}{j} \right) - \left( \sum_{k=0}^{a-1} \binom{k}{j} \right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof.  $\square$

**Lemma 3.7** (Centered hockey-stick identity). *For integers  $n \geq 1$ , and arbitrary integers  $t, k, r$*

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

*Proof.* By setting  $a = 1 - t + \frac{k}{2} + r$ , and  $b = n - t + \frac{k}{2} + r$  into Lemma (3.6) yields

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

$\square$

#### 4. ORDINARY SUMS OF POWERS

By decomposition of Newton's formula for powers (3.5), for integers  $n, m \geq 0$ , and an arbitrary integer  $t$ , we get

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \sum_{j=1}^n \left[ \binom{j-t+\frac{k}{2}}{k} + \binom{j-t+\frac{k}{2}-1}{k} \right] \delta^k t^m. \quad (1)$$

Now we reach the transition to closed form of ordinary sums of powers, by setting  $r = -1$ , and  $r = -2$  into Lemma (3.7). Therefore,

**Proposition 4.1** (Ordinary sums of powers decomposition). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[ \binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1} \right] \delta^k t^m.$$

*Proof.* We have the sums of powers expressed by (1), thus,

$$\sum_{j=1}^n \left[ \binom{j-t+\frac{k}{2}}{k} + \binom{j-t+\frac{k}{2}-1}{k} \right] = \binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1},$$

by setting  $r = -1$ , and  $r = -2$  into Lemma (3.7). □

For example,

**Example 4.2** (Sums of squares in central differences).

$$\begin{aligned} t = 0 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(1+n) + \frac{\delta^2 t^2}{2} \frac{n(1+3n+2n^2)}{6}, \\ t = 1 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-1) + \frac{\delta^2 t^2}{2} \frac{n(1-3n+2n^2)}{6}, \\ t = 2 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-3) + \frac{\delta^2 t^2}{2} \frac{n(13-9n+2n^2)}{6}, \\ t = 3 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-5) + \frac{\delta^2 t^2}{2} \frac{n(37-15n+2n^2)}{6}, \\ t = 4 : \quad \Sigma^1 n^2 &= \frac{t^2}{2}(2n) + \frac{\delta t^2}{2} n(n-7) + \frac{\delta^2 t^2}{2} \frac{n(73-21n+2n^2)}{6}. \end{aligned}$$

**Example 4.3** (Sums of cubes in central differences).

$$\begin{aligned} t = 0 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(1+n) + \frac{\delta^2 t^3}{2} \frac{n(1+3n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-1+n+4n^2+2n^3)}{24}, \\ t = 1 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n-1) + \frac{\delta^2 t^3}{2} \frac{n(1-3n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(1+n-4n^2+2n^3)}{24}, \\ t = 2 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n-3) + \frac{\delta^2 t^3}{2} \frac{n(13-9n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-21+25n-12n^2+2n^3)}{24}, \\ t = 3 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n-5) + \frac{\delta^2 t^3}{2} \frac{n(37-15n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-115+73n-20n^2+2n^3)}{24}, \\ t = 4 : \quad \Sigma^1 n^3 &= \frac{t^3}{2}(2n) + \frac{\delta t^3}{2} n(n-7) + \frac{\delta^2 t^3}{2} \frac{n(73-21n+2n^2)}{6} + \frac{\delta^3 t^3}{2} \frac{n(-329+145n-28n^2+2n^3)}{24}. \end{aligned}$$

It is indeed interesting to notice the connection between the decomposition of ordinary sums of powers (4.1) and Riordan's central factorial numbers of the second kind, that is

**Proposition 4.4** (Central factorial numbers sums). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[ \binom{n + \frac{k}{2} + 1}{k+1} - \binom{1 + \frac{k}{2}}{k+1} + \binom{n + \frac{k}{2}}{k+1} - \binom{\frac{k}{2}}{k+1} \right] k! T(m, k),$$

where  $T(n, k)$  denotes the central factorial numbers of the second kind (in Riordan's notation). See [5, p. 213], and [6].

These central factorial numbers of the second kind  $T(n, k)$  are defined by central Newton's formula for powers (3.2) evaluated in zero, such that

$$n^m = \sum_{k=0}^{\infty} T(m, k) n^{[k]}.$$

Which implies

$$T(n, k) = \frac{\delta^k 0^m}{k!},$$

by Newton's formula (3.2).

## 5. REMARK ON CENTRAL FACTORIAL NUMBERS OF THE SECOND KIND

In [1] Donald Knuth gives a remarkable formula for sums of odd powers

**Proposition 5.1** (Knuth's odd powers sum). *For non-negative integers  $n, m$*

$$\Sigma^1 n^{2m-1} = \sum_{k=1}^m (2k-1)! T(2m, 2k) \binom{n+k}{2k},$$

where  $T(2m, 2k)$  are central factorial numbers of the second kind.

It is quite interesting to notice that Proposition (5.1) originates from centered Newton's formula for powers (3.2) evaluated at zero. By Newton's formula, we have

$$n^{2m} = \sum_{k=0}^{2m} \frac{n^{[k]}}{k!} \cdot \delta^k 0^{2m}$$

The iteration for  $k = 0$  can be omitted because  $\delta^0 0^{2m} = 0$ . Next, we observe that  $\delta^k 0^{2m} = 0$  for every odd  $k$ , thus

$$n^{2m} = \sum_{k=1}^m \frac{n^{[2k]}}{(2k)!} \cdot \delta^{2k} 0^{2m}.$$

By definition of central factorial numbers of the second kind, yields

$$n^{2m} = \sum_{k=1}^m n^{[2k]} T(2m, 2k).$$

By applying identity (3.3), we get

$$\begin{aligned} n^{2m} &= \sum_{k=1}^m \frac{(2k)!}{(2k)!} n^{[2k]} T(2m, 2k) = \sum_{k=1}^m (2k)! \frac{n}{2k} \binom{n+k-1}{2k-1} T(2m, 2k) \\ &= \sum_{k=1}^m (2k-1)! n \binom{n+k-1}{2k-1} T(2m, 2k). \end{aligned}$$

By factoring out  $n$ , we obtain

$$n^{2m-1} = \sum_{k=1}^m (2k-1)! \binom{n+k-1}{2k-1} T(2m, 2k).$$

By hockey stick identity  $\sum_{k=0}^n \binom{k}{j} = \binom{n+1}{j+1}$  yields

$$\Sigma^1 n^{2m-1} = \sum_{k=1}^m (2k-1)! \binom{n+k}{2k} T(2m, 2k),$$

which is exactly the Proposition (5.1). Clearly, sum of odd powers above can be generalized to multifold case, by applying hockey stick pattern repeatedly

**Proposition 5.2** (Knuth's odd powers sum multifold). *For non-negative integers  $r, n, m$*

$$\Sigma^{r+1} n^{2m-1} = \sum_{k=1}^m (2k-1)! \binom{n+k+r}{2k+r} T(2m, 2k),$$

where  $T(2m, 2k)$  are central factorial numbers of the second kind.

## 6. GENERALIZED CENTRAL FACTORIAL NUMBERS OF THE SECOND KIND

**Definition 6.1** (Generalized central factorial numbers of the second kind). *For integers  $0 \leq k \leq m$ , and an arbitrary integer  $t$ , the generalized central factorial numbers of the second kind are defined by the relation*

$$n^m = \sum_{k=0}^{\infty} T_t(m, k) \cdot (n - t)^{[k]}.$$

It is also straightforward that

$$T_t(m, k) = \frac{\delta^k t^m}{k!},$$

by Newton's formula (3.2). For example,

| $n/k$ | 0 | 1              | 2 | 3               | 4  | 5              | 6  | 7 | 8 |
|-------|---|----------------|---|-----------------|----|----------------|----|---|---|
| 0     | 1 |                |   |                 |    |                |    |   |   |
| 1     | 0 | 1              |   |                 |    |                |    |   |   |
| 2     | 0 | 0              | 1 |                 |    |                |    |   |   |
| 3     | 0 | $\frac{1}{4}$  | 0 | 1               |    |                |    |   |   |
| 4     | 0 | 0              | 1 | 0               | 1  |                |    |   |   |
| 5     | 0 | $\frac{1}{16}$ | 0 | $\frac{5}{2}$   | 0  | 1              |    |   |   |
| 6     | 0 | 0              | 1 | 0               | 5  | 0              | 1  |   |   |
| 7     | 0 | $\frac{1}{64}$ | 0 | $\frac{91}{16}$ | 0  | $\frac{35}{4}$ | 0  | 1 |   |
| 8     | 0 | 0              | 1 | 0               | 21 | 0              | 14 | 0 | 1 |

**Table 1.** Values of generalized central factorial numbers  $T_0(n, k)$ . See also the sequences [A395862](#), and [A370703](#) in [7].



| $n/k$ | 0 | 1                 | 2   | 3                 | 4   | 5               | 6  | 7 | 8 |
|-------|---|-------------------|-----|-------------------|-----|-----------------|----|---|---|
| 0     | 1 |                   |     |                   |     |                 |    |   |   |
| 1     | 1 | 1                 |     |                   |     |                 |    |   |   |
| 2     | 1 | 2                 | 1   |                   |     |                 |    |   |   |
| 3     | 1 | $\frac{13}{4}$    | 3   | 1                 |     |                 |    |   |   |
| 4     | 1 | 5                 | 7   | 4                 | 1   |                 |    |   |   |
| 5     | 1 | $\frac{121}{16}$  | 15  | $\frac{25}{2}$    | 5   | 1               |    |   |   |
| 6     | 1 | $\frac{91}{8}$    | 31  | 35                | 20  | 6               | 1  |   |   |
| 7     | 1 | $\frac{1093}{64}$ | 63  | $\frac{1491}{16}$ | 70  | $\frac{119}{4}$ | 7  | 1 |   |
| 8     | 1 | $\frac{205}{8}$   | 127 | $\frac{483}{2}$   | 231 | 126             | 42 | 8 | 1 |

**Table 2.** Values of generalized central factorial numbers  $T_1(n, k)$ . See also the sequences [A395860](#), and [A395861](#) in [7].

| $n/k$ | 0   | 1                  | 2    | 3                  | 4    | 5               | 6   | 7  | 8 |
|-------|-----|--------------------|------|--------------------|------|-----------------|-----|----|---|
| 0     | 1   |                    |      |                    |      |                 |     |    |   |
| 1     | 2   | 1                  |      |                    |      |                 |     |    |   |
| 2     | 4   | 4                  | 1    |                    |      |                 |     |    |   |
| 3     | 8   | $\frac{49}{4}$     | 6    | 1                  |      |                 |     |    |   |
| 4     | 16  | 34                 | 25   | 8                  | 1    |                 |     |    |   |
| 5     | 32  | $\frac{1441}{16}$  | 90   | $\frac{85}{2}$     | 10   | 1               |     |    |   |
| 6     | 64  | $\frac{931}{4}$    | 301  | 190                | 65   | 12              | 1   |    |   |
| 7     | 128 | $\frac{37969}{64}$ | 966  | $\frac{12411}{16}$ | 350  | $\frac{371}{4}$ | 14  | 1  |   |
| 8     | 256 | $\frac{6001}{4}$   | 3025 | 3003               | 1701 | 588             | 126 | 16 | 1 |

**Table 3.** Values of generalized central factorial numbers  $T_2(n, k)$ . See also the sequences [A000000](#), and [A000000](#) in [7].

| $n/k$ | 0    | 1                   | 2     | 3                  | 4    | 5               | 6   | 7  | 8 |
|-------|------|---------------------|-------|--------------------|------|-----------------|-----|----|---|
| 0     | 1    |                     |       |                    |      |                 |     |    |   |
| 1     | 3    | 1                   |       |                    |      |                 |     |    |   |
| 2     | 9    | 6                   | 1     |                    |      |                 |     |    |   |
| 3     | 27   | $\frac{109}{4}$     | 9     | 1                  |      |                 |     |    |   |
| 4     | 81   | 111                 | 55    | 12                 | 1    |                 |     |    |   |
| 5     | 243  | $\frac{6841}{16}$   | 285   | $\frac{185}{2}$    | 15   | 1               |     |    |   |
| 6     | 729  | $\frac{12753}{8}$   | 1351  | 585                | 140  | 18              | 1   |    |   |
| 7     | 2187 | $\frac{372709}{64}$ | 6069  | $\frac{53011}{16}$ | 1050 | $\frac{791}{4}$ | 21  | 1  |   |
| 8     | 6561 | $\frac{167943}{8}$  | 26335 | $\frac{35049}{2}$  | 6951 | 1722            | 266 | 24 | 1 |

**Table 4.** Values of generalized central factorial numbers  $T_3(n, k)$ . See also the sequences [A000000](#), and [A000000](#) in [7].

| $t$ | $T_t(2m, 2k)$                                                                | OEIS ID                 |
|-----|------------------------------------------------------------------------------|-------------------------|
| 0   | 1, 0, 1, 0, 1, 1, 0, 1, 5, 1, 0, 1, 21, 14, 1, 0, 1, 85, 147, 30, 1, $\dots$ | <a href="#">A269945</a> |
| 1   | 1, 1, 1, 1, 7, 1, 1, 31, 20, 1, 1, 127, 231, 42, 1, 1, 511, $\dots$          | <a href="#">A394692</a> |
| 2   | 1, 4, 1, 16, 25, 1, 64, 301, 65, 1, 256, 3025, 1701, 126, 1, $\dots$         | <a href="#">A395456</a> |
| 3   | 1, 9, 1, 81, 55, 1, 729, 1351, 140, 1, 6561, 26335, 6951, 266, 1, $\dots$    | <a href="#">A395457</a> |

**Table 5.** Sequences of  $T_t(2m, 2k)$ .

Thus, we have the formula for sums of powers in terms of generalized central factorial numbers of the second kind.

**Proposition 6.2** (Generalized Central factorial sums). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\Sigma^1 n^m = \frac{1}{2} \sum_{k=0}^m \left[ \binom{n-t+\frac{k}{2}+1}{k+1} - \binom{1-t+\frac{k}{2}}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1} \right] k! T_t(m, k).$$

## 7. DOUBLE AND TRIPLE SUMS OF POWERS IN TERMS OF CENTRAL DIFFERENCES

**Proposition 7.1** (Double sums of powers decomposition). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\begin{aligned} \Sigma^2 n^m = \frac{1}{2} \sum_{k=0}^m \left[ + \binom{n-t+\frac{k}{2}+2}{k+2} - \binom{2-t+\frac{k}{2}}{k+2} \Sigma^0 n^0 - \binom{1-t+\frac{k}{2}}{k+1} \Sigma^1 n^0 \right. \\ \left. + \binom{n-t+\frac{k}{2}+1}{k+2} - \binom{1-t+\frac{k}{2}}{k+2} \Sigma^0 n^0 - \binom{0-t+\frac{k}{2}}{k+1} \Sigma^1 n^0 \right] \delta^k t^m \end{aligned}$$

For example,

**Example 7.2** (Double sum of squares in central differences).

$$\begin{aligned} t = 0 : \quad \Sigma^2 n^2 &= \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(1+n)^2(2+n)}{12}, \\ t = 1 : \quad \Sigma^2 n^2 &= \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-1+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n^2(-1+n^2)}{12}, \\ t = 2 : \quad \Sigma^2 n^2 &= \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-4-3n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(10+5n-4n^2+n^3)}{12}, \\ t = 3 : \quad \Sigma^2 n^2 &= \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-7-6n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(32+23n-8n^2+n^3)}{12}, \\ t = 4 : \quad \Sigma^2 n^2 &= \frac{t^2}{2} n(1+n) + \frac{\delta t^2}{2} \frac{n(-10-9n+n^2)}{3} + \frac{\delta^2 t^2}{2} \frac{n(66+53n-12n^2+n^3)}{12}. \end{aligned}$$

**Proposition 7.3** (Triple sums of powers decomposition). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\begin{aligned} \Sigma^3 n^m = \frac{1}{2} \sum_{k=0}^m \left[ + \binom{n-t+\frac{k}{2}+3}{k+3} - \binom{3-t+\frac{k}{2}}{k+3} \Sigma^0 n^0 - \binom{2-t+\frac{k}{2}}{k+2} \Sigma^1 n^0 - \binom{1-t+\frac{k}{2}}{k+1} \Sigma^2 n^0 \right. \\ \left. + \binom{n-t+\frac{k}{2}+2}{k+3} - \binom{2-t+\frac{k}{2}}{k+3} \Sigma^0 n^0 - \binom{1-t+\frac{k}{2}}{k+2} \Sigma^1 n^0 - \binom{0-t+\frac{k}{2}}{k+1} \Sigma^2 n^0 \right] \delta^k t^m \end{aligned}$$

For example,

**Example 7.4** (Triple sum of squares in central differences).

$$t = 0 : \quad \Sigma^3 n^2 = \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(6+11n+6n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(18+45n+40n^2+15n^3+2n^4)}{120},$$

$$t = 1 : \quad \Sigma^3 n^2 = \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-2-n+2n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(-2-5n+5n^3+2n^4)}{120},$$

$$t = 2 : \quad \Sigma^3 n^2 = \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-10-13n-2n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(58+65n-5n^3+2n^4)}{120},$$

$$t = 3 : \quad \Sigma^3 n^2 = \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-18-25n-6n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(198+255n+40n^2-15n^3+2n^4)}{120},$$

$$t = 4 : \quad \Sigma^3 n^2 = \frac{t^2}{2} \frac{n(2+3n+n^2)}{3} + \frac{\delta t^2}{2} \frac{n(-26-37n-10n^2+n^3)}{12} + \frac{\delta^2 t^2}{2} \frac{n(418+565n+120n^2-25n^3+2n^4)}{120}.$$

## 8. MULTIFOLD SUMS OF POWERS IN TERMS OF CENTRAL DIFFERENCES

**Proposition 8.1** (Multifold sums of powers). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\begin{aligned} \Sigma^r n^m &= \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right. \\ &\quad \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \Sigma^s n^0 \right\} \delta^k t^m. \end{aligned}$$

For example,

**Example 8.2** (Examples of multifold sums).

$$\begin{aligned}
t = 0 : \quad \Sigma^r n^m &= \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n+\frac{k}{2}+r}{k+r} + \binom{n+\frac{k}{2}+r-1}{k+r} \right. \\
&\quad \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s+\frac{k}{2}}{k+r-s} + \binom{r-1-s+\frac{k}{2}}{k+r-s} \right] \Sigma^s n^0 \right\} \delta^k 0^m, \\
t = 1 : \quad \Sigma^r n^m &= \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-1+\frac{k}{2}+r}{k+r} + \binom{n-1+\frac{k}{2}+r-1}{k+r} \right. \\
&\quad \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s-1+\frac{k}{2}}{k+r-s} + \binom{r-2-s+\frac{k}{2}}{k+r-s} \right] \Sigma^s n^0 \right\} \delta^k 1^m, \\
t = 2 : \quad \Sigma^r n^m &= \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-2+\frac{k}{2}+r}{k+r} + \binom{n-2+\frac{k}{2}+r-1}{k+r} \right. \\
&\quad \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s-2+\frac{k}{2}}{k+r-s} + \binom{r-3-s+\frac{k}{2}}{k+r-s} \right] \Sigma^s n^0 \right\} \delta^k 2^m.
\end{aligned}$$

## 9. MULTIFOLD SUMS OF POWERS IN GENERALIZED CENTRAL FACTORIAL NUMBERS

**Proposition 9.1** (Multifold central factorial sums of powers). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\begin{aligned}
\Sigma^r n^m &= \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right. \\
&\quad \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \Sigma^s n^0 \right\} k! T_t(m, k).
\end{aligned}$$

**Example 9.2** (Sums of squares in central factorial numbers).

$$t = 0 : \quad \Sigma^1 n^2 = n \cdot 0 + \frac{1}{2}n(1+n) \cdot 0 + \frac{1}{6}n(1+3n+2n^2) \cdot 1,$$

$$t = 1 : \quad \Sigma^1 n^2 = n \cdot 1 + \frac{1}{2}n(n-1) \cdot 2 + \frac{1}{6}n(1-3n+2n^2) \cdot 1,$$

$$t = 2 : \quad \Sigma^1 n^2 = n \cdot 4 + \frac{1}{2}n(n-3) \cdot 4 + \frac{1}{6}n(13-9n+2n^2) \cdot 1,$$

$$t = 3 : \quad \Sigma^1 n^2 = n \cdot 9 + \frac{1}{2}n(n-5) \cdot 6 + \frac{1}{6}n(37-15n+2n^2) \cdot 1,$$

$$t = 4 : \quad \Sigma^1 n^2 = n \cdot 16 + \frac{1}{2}n(n-7) \cdot 8 + \frac{1}{6}n(73-21n+2n^2) \cdot 1,$$

$$t = 5 : \quad \Sigma^1 n^2 = n \cdot 25 + \frac{1}{2}n(n-9) \cdot 10 + \frac{1}{6}n(121-27n+2n^2) \cdot 1.$$

From the example above, we may observe that

- For  $t = 0$  the coefficients 0, 0, 1 correspond to the second row of the triangle (1).
- For  $t = 1$  the coefficients 1, 2, 1 correspond to the second row of the triangle (2).
- For  $t = 2$  the coefficients 4, 4, 1 correspond to the second row of the triangle (3).
- For  $t = 3$  the coefficients 9, 6, 1 correspond to the second row of the triangle (4).

**Example 9.3** (Sums of cubes in central factorial numbers).

$$\begin{aligned}
t = 0 : \quad \Sigma^1 n^4 &= n \cdot 0 + \frac{1}{2}n(1+n) \cdot 0 + \frac{1}{6}n(1+3n+2n^2) \cdot 1 \\
&\quad + \frac{1}{8}n(-1+n+4n^2+2n^3) \cdot 0 + \frac{1}{10}n(-2-5n+5n^3+2n^4) \cdot 1, \\
t = 1 : \quad \Sigma^1 n^4 &= n \cdot 1 + \frac{1}{2}n(n-1) \cdot 5 + \frac{1}{6}n(1-3n+2n^2) \cdot 7 \\
&\quad + \frac{1}{8}n(1+n-4n^2+2n^3) \cdot 4 + \frac{1}{10}n(-2+5n-5n^3+2n^4) \cdot 1, \\
t = 2 : \quad \Sigma^1 n^4 &= n \cdot 16 + \frac{1}{2}n(n-3) \cdot 34 + \frac{1}{6}n(13-9n+2n^2) \cdot 25 \\
&\quad + \frac{1}{8}n(-21+25n-12n^2+2n^3) \cdot 8 \\
&\quad + \frac{1}{10}n(18-45n+40n^2-15n^3+2n^4) \cdot 1, \\
t = 3 : \quad \Sigma^1 n^4 &= n \cdot 81 + \frac{1}{2}n(n-5) \cdot 111 + \frac{1}{6}n(37-15n+2n^2) \cdot 55 \\
&\quad + \frac{1}{8}n(-115+73n-20n^2+2n^3) \cdot 12 \\
&\quad + \frac{1}{10}n(298-275n+120n^2-25n^3+2n^4) \cdot 1
\end{aligned}$$

From the example above, we may observe that

- For  $t = 0$  the coefficients 0, 0, 1, 0, 1 correspond to the fourth row of the triangle (1).
- For  $t = 1$  the coefficients 1, 5, 7, 4, 1 correspond to the fourth row of the triangle (2).
- For  $t = 2$  the coefficients 16, 34, 25, 8, 1 correspond to the fourth row of the triangle (3).
- For  $t = 3$  the coefficients 81, 111, 55, 12, 1 correspond to the fourth row of the triangle (4).

## 10. MULTIFOLD SUMS OF POWERS IN BINOMIAL FORM

**Lemma 10.1** (Multifold sum of zero powers). *For integers  $r \geq 0$  and  $n \geq 1$*

$$\Sigma^r n^0 = \binom{r+n-1}{r}$$

*Proof.* (1) Let  $r = 0$ , then we have  $\Sigma^0 n^0 = 1$ , by definition (1.1).

- (2) Let  $r = 1$ , then we have  $\Sigma^1 n^0 = \sum_{k=1}^n 1 = \binom{n}{1}$ .
- (3) Let  $r = 2$ , then we have  $\Sigma^2 n^0 = \sum_{k=1}^n \binom{k}{1} = \binom{n+1}{2}$ .
- (4) Let  $r = 3$ , then we have  $\Sigma^3 n^0 = \sum_{k=1}^n \binom{k+1}{2} = \binom{n+2}{3}$ .
- (5) Similarly, for arbitrary integer  $r$ , by hockey stick identity  $\sum_{k=1}^n \binom{k}{r} = \binom{n+1}{r+1}$ , yields
- $$\Sigma^r n^0 = \sum_{k=1}^n \Sigma^{r-1} k^0 = \sum_{k=1}^n \binom{k+r-2}{r-1} = \binom{r+n-1}{r}.$$

□

**Proposition 10.2** (Multifold binomial sums of powers). *For integers  $n, m \geq 0$ , and an arbitrary integer  $t$ ,*

$$\begin{aligned} \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s-t+\frac{k}{2}}{k+r-s} + \binom{r-1-s-t+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k t^m \end{aligned}$$

For example,

**Example 10.3** (Examples of multifold binomial sums).

$$\begin{aligned} t = 0 : \quad \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n+\frac{k}{2}+r}{k+r} + \binom{n+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s+\frac{k}{2}}{k+r-s} + \binom{r-1-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k 0^m, \\ t = 1 : \quad \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-1+\frac{k}{2}+r}{k+r} + \binom{n-1+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s-1+\frac{k}{2}}{k+r-s} + \binom{r-2-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k 1^m, \\ t = 2 : \quad \Sigma^r n^m = & \frac{1}{2} \sum_{k=0}^m \left\{ \binom{n-2+\frac{k}{2}+r}{k+r} + \binom{n-2+\frac{k}{2}+r-1}{k+r} \right. \\ & \left. - \sum_{s=0}^{r-1} \left[ \binom{r-s-2+\frac{k}{2}}{k+r-s} + \binom{r-3-s+\frac{k}{2}}{k+r-s} \right] \binom{s+n-1}{s} \right\} \delta^k 2^m. \end{aligned}$$



## 11. RELATED WORKS

## 12. FRAMEWORK FOR SUMS OF POWERS

## 13. CONCLUSIONS

## REFERENCES

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## 14. MATHEMATICA PROGRAMS

Use this [GitHub](#) repository to validate the results using Mathematica programs.

| Mathematica Function                                        | Validates / Prints              |
|-------------------------------------------------------------|---------------------------------|
| <code>MultifoldSumOfPowersRecurrence[r, n, m]</code>        | Computes $\Sigma^r n^m$ , (1.1) |
| <code>CentralFactorial[n, k]</code>                         | Computes $n^{[k]}$ , (1.2)      |
| <code>CentralFactorialsBinomialFormDecomposition[20]</code> | Validates Lem. (3.4)            |
| <code>ValidateNewtonsFormulaForPowers[10]</code>            | Validates Lem. (3.5)            |
| <code>ValidateCenteredHockeyStickIdentity[5]</code>         | Validates Lem. (3.7)            |
| <code>ValidateOrdinarySumsOfPowersDecomposition[10]</code>  | Validates Prop. (4.1)           |
| <code>ValidateDoubleSumsOfPowersDecomposition[5]</code>     | Validates Prop. (7.1)           |
| <code>ValidateTripleSumsOfPowersDecomposition[5]</code>     | Validates Prop. (7.3)           |
| <code>ValidateMultifoldSumsOfPowers[3]</code>               | Validates Prop. (8.1)           |